Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Author of an IEEE conference publication with hands-on experience developing end-to-end machine learning solutions using Python, PyTorch, TensorFlow, Scikit-learn, FastAPI, Docker, and AWS. Experienced in building end-to-end machine learning pipelines including data preprocessing, feature engineering, model training, evaluation, deployment, and NLP applications. Strong problem-solving skills with a passion for developing production-ready AI systems.

SKILLS

- **Programming :** Python, SQL, C++, Bash
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-Learn, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Model Evaluation, Feature Engineering, Data Preprocessing
- **Data Analysis:** NumPy, Pandas, Matplotlib, Seaborn
- **Backend & APIs :** FastAPI, REST APIs
- **Cloud & DevOps :** Docker, AWS, ECS, CloudWatch, Git, Linux, Model Deployment, Containerization

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 2025**

- Developed reproducible machine learning pipelines for data preprocessing, model training, evaluation, and benchmarking using Python and PyTorch.
- Evaluated Vision Transformer models under continual learning settings using standardized performance metrics.
- Automated experiment execution, performance tracking, and comparative analysis across multiple learning strategies.
- Performed hyperparameter tuning and comparative performance evaluation to improve model accuracy and robustness.
- Documented experimental results and identified improvements for model performance and knowledge retention.
- **Technologies:** Python, PyTorch, Vision Transformers, Git

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Analyzed Random Forest, SVM, Genetic Algorithm, LS-SVM, Decision Trees, and Artificial Neural Networks across multiple air quality datasets.
- Evaluated multiple machine learning models using standard performance metrics to identify the most effective approach for air quality prediction.
- Presented scientific findings through structured research documentation and publication.

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence platform for semantic document comparison using Sentence Transformers and vector embeddings.
- Built end-to-end NLP pipelines including document preprocessing, parsing, chunking, embedding generation, and semantic similarity analysis.

- Developed FastAPI REST APIs to expose AI-powered document comparison services.
- Implemented embedding-based semantic retrieval and document change detection using Sentence Transformers and cosine similarity.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured CloudWatch monitoring to track application health and performance.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS, Git

Metal Defect Detection using Computer Vision

- Developed an end-to-end computer vision pipeline for automated metal surface defect detection using the NEU Metal Surface Defects dataset.
- Performed image preprocessing, data augmentation, feature extraction, and dataset preparation to improve model training and classification performance.
- Built, trained, and evaluated CNN, Support Vector Machine (SVM), and Random Forest models using TensorFlow, Keras, and Scikit-learn.
- Conducted systematic model evaluation using accuracy, precision, recall, F1-score, confusion matrices, and class-wise performance analysis across multiple defect categories.
- Compared deep learning and classical machine learning models to evaluate accuracy, efficiency, and suitability for industrial defect detection.
- Optimized preprocessing and feature extraction techniques to improve classification performance across multiple defect classes.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn, NumPy, Git

Lossy Text Compressor [Semantic NLP Pipeline]

- Developed a semantic text compression pipeline using transformer embeddings, quantization, and similarity-based reconstruction techniques.
- Implemented embedding generation, semantic reconstruction, and similarity evaluation workflows.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics across multiple quantization levels.
- Analyzed trade-offs between compression efficiency and semantic preservation to identify optimal compression settings.
- **Technologies:** Python, PyTorch, Transformer Models, Git